Principles of Acoustics and **Vibration**

Single DoF Oscillator (Unforced) - Mass on a Spring

Joshua Meggitt

Acoustics Research Centre, University of Salford

Who am I?

- Acoustic Research Centre, University of Salford:
  - BEng 2011-2014
  - PhD 2014-2017
  - Post Doc 2017-2019
  - Lecturer 2019-present
- PhD on in-situ characterisation of vibration isolators, virtual acoustic prototyping
- Research interests:
  - Anything and everything blocked force
  - Hybrid methods for vibro-acoustic simulation
  - Uncertainty in TPA methods
  - A little bit of acoustics
- Non-research interests:
  - Trying to skateboard
  - Brazilian Jiu Jitsu

A bit of context

- What does acoustics and vibration have in common?
- Understanding simple harmonic oscillation is of fundamental importance; principles apply across all aspects of acoustics.
- We will introduce concepts through study of Single Degree of Freedom (DoF) mass-spring system.
- Will provide us with a gentle introduction for those with mathematical cobwebs, but also introduce us to concepts that will translate over to systems of much greater complexity.
- Module topics we will cover:
  1. Single DoF mass-spring - part 1 (unforced response)
  2. Single DoF mass-spring - part 2 (forced response)
  3. Two DoF mass-spring system
  4. N-DoF systems
  5. Discrete to continuous systems.

Today's plan

1. Equation of motion
2. Coordinates
3. Undamped homogenous solution
4. Damped homogenous solution
5. Estimating damping

Equation of motion

A bit of context

- Mass-spring-damper - the simplest mechanical system of some practical interest...
- We want to derive an equation that describes the motion of the mass, given some IC and/or externall applied forcing.
- Newton's second law states that,

$$F = ma \qquad \sum_i F_i = m\ddot{x} \qquad (1)$$

- What forces are acting on our mass - **free body diagram**.

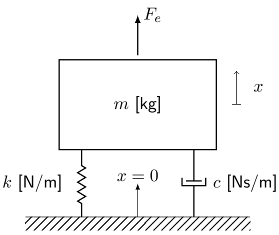

Figure: Mass-spring-damper system

Mass-spring-damper system

- Newton's second law states that,

$$m\ddot{x} = \sum_i F_i$$

$$m\ddot{x} = +F_e - F_k - F_c \qquad (2)$$

- F_e is an external force, defined in the direction of positive travel.
- F_k and F_c are forces that oppose motion - but what are they?

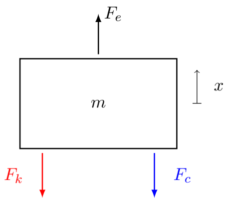

Figure: Mass-spring-damper system

Ideal springs - Hooke's law

- An ideal spring has no mass and generates a force proportion to its elongation Δx .
- Force in opposite direction to displacement and linearly related to the elongation of the spring,

$$F_k = -kx_1 \qquad (3)$$

- k is called the *spring stiffness* (N/m).
- For a two-sided spring the force is equal and opposite in direction,

$$F_k = k\overbrace{(x_2 - x_1)}^{\Delta x} \qquad (4)$$

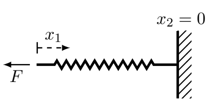

Figure: Fixed linear spring.

Figure: Ideal linear spring.

Ideal springs - linearity

- Spring force always opposes displacement.
  - If $\Delta x < 0$ then $F_k > 0$
  - If $\Delta x > 0$ then $F_k < 0$
- Ideal springs are linear - the spring force and elongation remain linearly related indefinitely,

$$F_k = -kx_1 \qquad 2F_k = -k\,2x_1 \tag{5}$$

- Real springs are non-linear:
  - The force/displacement relation begins to change for large displacements.
  - Softening or hardening springs
- What about combinations of springs? Two options: series or parallel

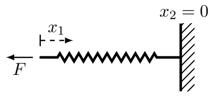

Figure: Fixed linear spring.

Figure: Ideal linear spring.

Ideal springs - parallel combination

- In parallel spring have the same displacement at either end.
- Force divides itself into F_{k1} and F_{k2},

$$F_{k1} = k_1(x_2 - x_1) \tag{6}$$
$$F_{k2} = k_2(x_2 - x_1) \tag{7}$$

- For adds to total force,

$$F_k = F_{k1} + F_{k2} \tag{8}$$
$$= \underbrace{(k_1 + k_2)}_{k_t}(x_2 - x_1) \tag{9}$$

- Total stiffness = sum of individual stiffnesses.

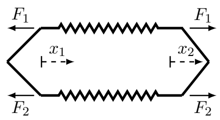

Figure: Parallel combined springs.

$$k_t = \sum_i^N k_i \tag{10}$$

Ideal springs - series combination

- In series springs have the same force through their lengths.

$$F_k = k_1(x_2 - x_1) \tag{11}$$
$$F_k = k_2(x_3 - x_2) \tag{12}$$

- From 11 we get,

$$x_2 = \frac{F_k}{k_1} + x_1 \tag{13}$$

- Substitute into 12,

$$F_k = k_2(x_3 - x_1 - \frac{F_k}{k_1}) \tag{14}$$

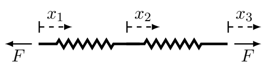

Figure: Series combined springs.

Ideal springs - series combination

- Substitute into 12,

$$F_k = k_2(x_3 - x_1 - \frac{F_k}{k_1}) \tag{15}$$

- A little bit of algebra leads to,

$$F_k = \underbrace{\left(\frac{1}{k_1} + \frac{1}{k_2}\right)^{-1}}_{k}(x_3 - x_1) \tag{16}$$

- Total stiffness always less than individual stiffnesses.

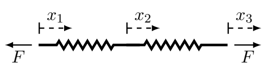

Figure: Series combined springs.

$$k = \left(\sum_i^N \frac{1}{k_i}\right) \tag{17}$$

Ideal damper - viscous dashpot

- An viscous damper has no mass and generates a force proportion to the relative velocity $\Delta\dot{x}$.
- Force in opposite direction and linearly related to the relative velocity of the dashpot,

$$F_c = -c\dot{x}_1 \tag{18}$$

- c is the *viscous damping coefficient* (Ns/m).
- For a two-sided dashpot,

$$F_c = c\,\overbrace{(\dot{x}_2 - \dot{x}_1)}^{\Delta\dot{x}} \tag{19}$$

- Same equations apply for series and parallel arrangements as for springs.

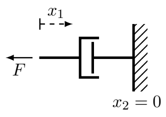

Figure: Fixed linear dashpot.

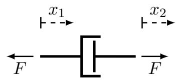

Figure: Ideal linear spring.

Spring-dashpot equation of motion

- Consider spring-dashpot with *no mass*.
- Newton's second law says,

$$F_e + F_k + F_c = 0 \tag{20}$$

- Substitute element forces into the above,

$$F_e - kx - c\dot{x} = 0 \tag{21}$$

- Or as a 'force balance',

$$F_e = kx + c\dot{x} \tag{22}$$

- Equation 22 is a first order linear differential equation with constant coefficients.

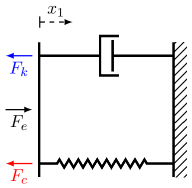

Figure: Fixed spring-dashpot system.

Mass-spring-dashpot EoM

- Newton's second law states that,

$$\begin{aligned} m\ddot{x} &= F_e + F_k + F_c \\ &= F_e - kx - c\dot{x} \end{aligned} \tag{23}$$

- Isolate external force,

$$F = m\ddot{x} + kx + c\dot{x} \tag{24}$$

- *Second* order linear differential equation with constant coefficients.
- Called *single degree of freedom (DoF)* system
  - DoF: number of independent coordinates required to completely describe configuration.

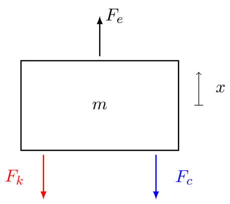

Figure: Mass-spring-damper system

Coordinates

Preferred coordinates

- Consider a the same problem, now including the effect of gravity.
  - y is the position of the mass relative to its static position y_0 ignoring gravity.
  - x is the position of the mass relative to its static position under gravity.
  - $x_{st} = y_0 - x_0$ is the static displacement due to gravity $(= mg/k)$
- EoM with respect to y_0,

$$F_e = m\ddot{y} + c\dot{y} + ky + mg \qquad (25)$$

- Extra term mg because y_0 is not at equilibrium x_0, EoM more complex.

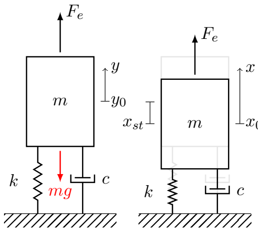

Figure: Mass-spring-damper system under gravity.

Preferred coordinates

- Writing the displacement as $y = x - x_{st}$ (recall that x is relative to x_0) the EoM becomes,

$$F_e = m\ddot{x} + c\dot{x} + k(x - x_{st}) + mg \qquad (26)$$

- Since $kx_{st} = mg$, EoM reduces to,

$$F_e = m\ddot{x} + c\dot{x} + kx \qquad (27)$$

- By using the equilibrium point x_0 as our reference point, our EoM gets simpler.
- More generally - *a change of coordinates can be used to simplify the problem!*
  - This will be very important once we move onto multi-DoF systems.

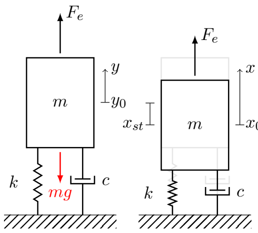

Figure: Mass-spring-damper system under gravity.

Undamped homogenous solution

Homogenous spring-dashpot solution

- Two types of solution to EoM: homogenous (unforced) and particular (forced)
- General solution is obtained as the sum of homogenous and particular.
- *Unforced* spring-dashpot EoM:

$$0 = kx + c\dot{x} \qquad (28)$$

- We are interested in solutions of the form $x(t) = Ae^{st}$. Substitute this into above.

$$0 = kAe^{st} + c\underbrace{sAe^{st}}_{\dot{x}} \qquad (29)$$

- Dividing both sides by Ae^{st} we obtain so-called *characteristic equation*,

$$0 = k + sc \qquad s = -\frac{k}{c} \qquad (30)$$

Homogenous spring-dashpot solution

- Characteristic equation from before,

$$0 = k + sc \qquad s = -\frac{k}{c} \tag{31}$$

- Using the above, the homogenous solution is $x(t) = Ae^{st} = Ae^{-\frac{k}{c}t} = Ae^{-\frac{t}{\tau}}$, where $\tau = c/k$ is called the 'time constant'.
- Suppose at $t = 0$ the system is displaced $x(0) = x_0$, we see that,

$$x(0) = Ae^{-\frac{0}{\tau}} = A = x_0 \tag{32}$$

- Solution due to initial displacement is then,

$$x(t) = x_0 e^{-\frac{t}{\tau}} \qquad t > 0 \tag{33}$$
$$x(t) = 0 \qquad t < 0 \tag{34}$$

Homogenous spring-dashpot solution

- Solution due to initial displacement is then,

$$x(t) = x_0 e^{-\frac{t}{\tau}} \qquad t > 0 \tag{35}$$
$$x(t) = 0 \qquad t < 0 \tag{36}$$

- Time constant controls speed at which system returns to equilibrium, $x = 0$.
  - Big $\tau \rightarrow$ slow return.
  - Small $\tau \rightarrow$ fast return.
- System returns to equilibrium with *no oscillation*.

Homogenous spring-dashpot solution

- Solution due to initial displacement is then,

$$x(t) = x_0 e^{-\frac{t}{\tau}} \qquad t > 0 \tag{37}$$
$$x(t) = 0 \qquad t < 0 \tag{38}$$

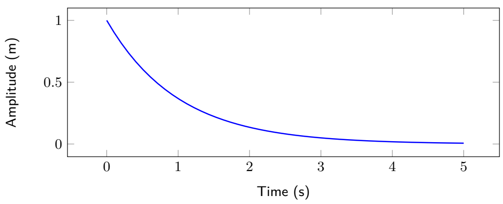

Figure: Displacement response due to initial displacement $x_0 = 1$.

Undamped homogenous mass-spring solution

- Undamped homogenous mass-spring EoM,

$$0 = m\ddot{x} + kx \qquad (F_e = c = 0) \tag{39}$$

- Divide through by m to get,

$$0 = \ddot{x} + \frac{k}{m}x = \ddot{x} + \omega_n^2 x \tag{40}$$

- Consider solution of the form $x(t) = Ae^{st}$,

$$0 = s^2 Ae^{st} + \omega_n^2 Ae^{st} \tag{41}$$

- Divide through by Ae^{st} (charac. eq.),

$$0 = s^2 + \omega_n^2 \qquad s^2 = -\omega_n^2 \tag{42}$$

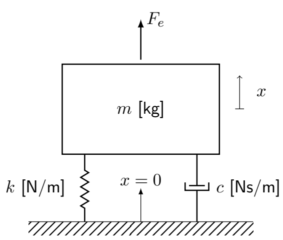

Figure: Mass-spring-damper system

Undamped homogenous mass-spring solution

- Characteristic equation,

$$0 = s^2 + \omega_n^2 \qquad s^2 = -\omega_n^2 \tag{43}$$

- Square root of both sides ($i = \sqrt{-1}$),

$$s = \pm\sqrt{-\omega_n^2} = \pm i\omega_n \tag{44}$$

- There are two roots, so the solution is the sum of each,

$$x(t) = A_1 e^{i\omega_n t} + A_2 e^{-i\omega_n t} \tag{45}$$

- A_1 and A_2 are constants that depend on initial conditions, x_0 and $\dot{x}_0$.
- Imaginary roots lead to complex exponentials - *harmonic motion at frequency* ω_n.
  - We call ω_n the natural frequency of vibration, or characteristic frequency.
  - Force free undamped system will *always* oscillate at ω_n, regardless of IC.

Undamped homogenous mass-spring solution

- Undamped homogenous solution, with $\omega_n = \sqrt{k/m}$,

$$x(t) = A_1 e^{i\omega_n t} + A_2 e^{-i\omega_n t} \tag{46}$$

- Solve for the constants A_1 and A_2 in terms of initial displacement and velocity.

$$x(0) = x_0 = A_1 + A_2 \tag{47}$$

$$\dot{x}(0) = \dot{x}_0 = i\omega_n A_1 - i\omega_n A_2 \quad \rightarrow \quad \frac{\dot{x}_0}{i\omega_n} = A_1 - A_2 \tag{48}$$

- From the above we obtain,

$$A_1 = \frac{1}{2}\left(x_0 + \frac{\dot{x}_0}{i\omega}\right) \qquad A_2 = \frac{1}{2}\left(x_0 - \frac{\dot{x}_0}{i\omega}\right) \tag{49}$$

- Notice that A_1 andA_2 are complex conjugates... So only two independent constants since $A_1 = a + ib$ and $A_2 = a - ib$.
  - This is expected as system is second order - two only two arbitrary constants.

Undamped homogenous mass-spring solution

- Substitute the constants into solution and use Euler's formula,

$$x(t) = \frac{1}{2}\left(x_0 + \frac{\dot{x}}{i\omega}\right)e^{i\omega_n t} + \frac{1}{2}\left(x_0 - \frac{\dot{x}}{i\omega}\right)e^{-i\omega_n t} \tag{50}$$

$$= \frac{1}{2}x_0 \underbrace{\left(e^{i\omega_n t} + e^{-i\omega_n t}\right)}_{2\cos\omega_n t} + \frac{1}{2}\frac{\dot{x}}{\omega}\frac{1}{i}\underbrace{\left(e^{i\omega_n t} - e^{-i\omega_n t}\right)}_{2\sin\omega_n t} \tag{51}$$

- From the above, the *real* solution is given by,

$$x(t) = x_0 \cos\omega_n t + \frac{\dot{x}_0}{\omega_n}\sin\omega_n t \tag{52}$$

- Satisfying initial conditions which are both real, has caused the imaginary part of $x(t)$ to vanish. Expected, true displacement must be real.
- We don't have to go through IC steps to make imaginary part disappear...

Undamped homogenous mass-spring solution

- Alternatively, we can use the solution in its original complex form, $x(t) = Ae^{i\omega_n t}$, where $A = a + ib$, and take the real part of the response,

$$\text{Real}[x(t)] = \text{Real}[(a + ib)(\cos\omega_n t + i\sin\omega_n t)] \tag{53}$$

$$= a\cos\omega_n t - b\sin\omega_n t \tag{54}$$

- This can be written equivalently as,

$$\text{Real}[x(t)] = |A|\cos(\omega_n t + \phi) \tag{55}$$

  where $|A| = \sqrt{a^2 + b^2}$ and $\phi = \tan^{-1}(b/a)$ are the amplitude and phase, respectively.
- We can find $|A|$ and ϕ directly from IC.

Undamped homogenous mass-spring solution

- To find $|A|$ and ϕ from IC consider solution at $t = 0$,

$$x(0) = x_0 = |A|\cos(\phi) \quad (56)$$
$$\dot{x}(0) = \dot{x}_0 = -|A|\omega_n \sin(\phi) \quad (57)$$

- Pair of simultaneous equations - solve for ϕ,

$$\frac{\dot{x}_0}{x_0} = -\omega_n \frac{\sin\phi}{\cos\phi} = -\omega_n \tan\phi \quad \rightarrow \quad \phi = \tan^{-1}\left(\frac{1}{\omega_n}\frac{\dot{x}_0}{x_0}\right) \quad (58)$$

- Square and sum x_0 and $\dot{x}_0/\omega_n$, then solve for $|A|$,

$$x_0^2 + \frac{\dot{x}_0^2}{\omega_n^2} = |A|^2\cos^2(\phi) + |A|^2\sin^2(\phi) = |A|^2 \overbrace{(\cos^2(\phi) + \sin^2(\phi))}^{=1} \quad (59)$$

- Amplitude becomes,

$$|A| = \sqrt{x_0^2 + \frac{\dot{x}_0^2}{\omega_n^2}} \quad (60)$$

Undamped homogenous mass-spring solution

- Solution of system to arbitrary IC can be written in several ways.
- In complex exponential form,

$$x(t) = A_1 e^{i\omega_n t} + A_2 e^{-i\omega_n t} \quad \text{with} \quad A_1 = \frac{1}{2}\left(x_0 + \frac{\dot{x}_0}{i\omega}\right) \quad A_2 = \frac{1}{2}\left(x_0 - \frac{\dot{x}_0}{i\omega}\right) \quad (61)$$

$$x(t) = \text{Real}[Ae^{i\omega_n t}] \quad \text{with} \quad A = x_0 + i\frac{\dot{x}_0}{\omega_n} \quad (62)$$

- In explicit $\cos(\quad)$ form,

$$x(t) = |A|\cos(\omega_n t - \phi) \quad \text{with} \quad A = \sqrt{x_0^2 + \frac{\dot{x}_0^2}{\omega_n^2}} \quad \phi = \tan^{-1}\left(\frac{1}{\omega_n}\frac{\dot{x}_0}{x_0}\right) \quad (63)$$

- Same result! Just different maths...

Damped homogenous solution

Damped homogenous mass-spring solution

- Damped homogenous mass-spring EoM,

$$0 = m\ddot{x} + kx + c\dot{x} \qquad (F_e = 0) \quad (64)$$

- Divide through by m to get,

$$0 = \ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x \quad (65)$$

using the defined viscous damping factor,

$$\zeta = \frac{c}{2m\omega_n} \quad (66)$$

- Assume solution of form $x(t) = Ae^{st}$,

$$0 = s^2Ae^{st} + 2\zeta\omega_n sAe^{st} + \omega_n^2 Ae^{st} \quad (67)$$

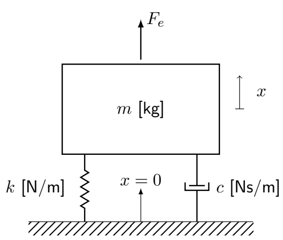

Figure: Mass-spring-damper system

Undamped homogenous mass-spring solution

- Divide through by Ae^{st},

$$0 = s^2 + 2\zeta\omega_n s + \omega_n^2 \tag{68}$$

- Quadratic in s, solve using quadratic formula (we defined ζ to make this equation nice!),

$$\begin{matrix} s_1 \\ s_2 \end{matrix} = \left(-\zeta \pm \sqrt{\zeta^2 - 1}\right)\omega_n \tag{69}$$

- The nature of the solution depends on the value of ζ!
  - If $\zeta = 0$ then $s_{1,2}$ are purely imaginary and the solution is that of undamped case.
  - If $0 < \zeta < 1$ then $s_{1,2}$ are complex (because of negative in square root) and the solution will become a decaying sinusoid.
  - If $\zeta = 1$ then $s_{1,2}$ are purely real and the solution decays exponentially at the fastest possible rate.
  - If $\zeta > 1$ then $s_{1,2}$ are purely real and the solution decays exponentially at an increasing slow rate.

Undamped homogenous mass-spring solution

- Plug $s_{1,2}$ into our solution $x(t) = Ae^{st}$,

$$x(t) = A_1 e^{\left(-\zeta+\sqrt{\zeta^2-1}\right)\omega_n t} + A_2 e^{\left(-\zeta+\sqrt{\zeta^2-1}\right)\omega_n t} \tag{70}$$

- We can factor out constant part of exponential $e^{-\zeta\omega_n t}$

$$x(t) = \left(A_1 e^{\sqrt{\zeta^2-1}\omega_n t} + A_2 e^{-\sqrt{\zeta^2-1}\omega_n t}\right) e^{-\zeta\omega_n t} \tag{71}$$

- If $\zeta > 1$ (so-called *over-damped*) $x(t)$ decays exponentially, with the precise shaper controlled by A_1 and A_2.
- If $\zeta = 1$ (so-called *critical damping*) can be shown that solution has the form,

$$x(t) = (A_1 + tA_2)e^{-\omega_n t} \tag{72}$$

- This occurs when,

$$\zeta = 1 = \frac{c}{2m\omega_n} \to c_{cr} = 2m\omega_n = 2\sqrt{km} \tag{73}$$

Undamped homogenous mass-spring solution

- More interesting is the so-called *under-damped case*, $0 < \zeta < 1$. In this case it is useful to rewrite $\sqrt{\zeta^2 - 1}$ as $i\sqrt{1-\zeta^2}$,

$$x(t) = \left(A_1 e^{i\sqrt{1-\zeta^2}\omega_n t} + A_2 e^{-i\sqrt{1-\zeta^2}\omega_n t}\right) e^{-\zeta\omega_n t} \tag{74}$$

- We define the *frequency of damped natural vibration*, $\omega_d = \sqrt{1-\zeta^2}\omega_n$.
  - The factor of $\sqrt{1-\zeta^2}$ slightly lowers the system's natural frequency.
  - Exponential decay remains proportional to the undamped natural frequency ω_n.
- Using damped natural frequency, our solution has the form,

$$x(t) = \left(A_1 e^{i\omega_d t} + A_2 e^{-i\omega_d t}\right) e^{-\zeta\omega_n t} \tag{75}$$

- Note that the bracketed term is identical in form to what we had for the undamped case, just with $\omega_n \to \omega_d$.

Undamped homogenous mass-spring solution

- Reusing the previous *undamped* result, our *damped* solution becomes,

$$x(t) = A\cos(\omega_d t - \phi)e^{-\zeta\omega_n t} \tag{76}$$

with the initial conditions,

$$A = \sqrt{x_0^2 + \frac{\dot{x}_0^2}{\omega_d^2}} \quad \text{and} \quad \phi = \tan^{-1}\left(\frac{1}{\omega_d}\frac{\dot{x}_0}{x_0}\right) \tag{77}$$

- We can start an oscillation with an initial displacement and/or velocity. How else might we push the system into motion?
  - We might apply a very short force input, i.e. give it a kick...

Undamped homogenous mass-spring solution

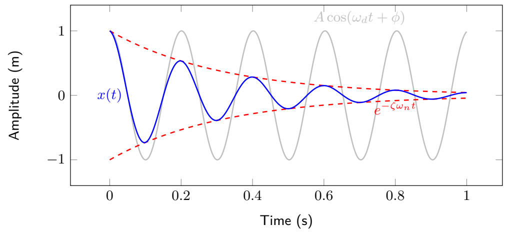

Figure: Under-damped system. IC: $x_0 = 1$, $\dot{x}_0 = 0$

Introduction Equation of motion Coordinates Undamped homogenous solution Damped homogenous solution Estimating damping Summary

Undamped homogenous mass-spring solution

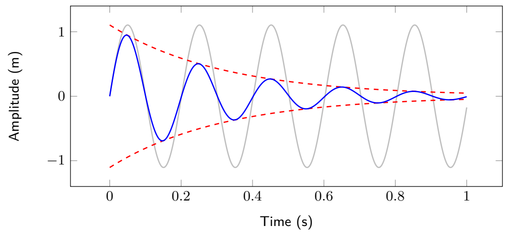

Figure: Under-damped system. IC: $x_0 = 0$, $\dot{x}_0 = 1e-4$

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Unit impulse response

- Rather than setting mass into motion by displacing or applying an initial velocity, suppose instead we give it a *kick*.
- This kick can be modelled as a constant force F over the infinitesimal time $0 < t < \epsilon$.
- During this kick, the velocity of the mass increases from 0 to $\dot{x}(\epsilon)$, after which its motion is governed by the system dynamics.
- From Newton's second law $F = m\ddot{x}$, and simple kinematics $\dot{x}(t) = t\ddot{x}$, we have that the velocity at $t = \epsilon$ is,

$$\dot{x}(\epsilon) = \epsilon\ddot{x} = \frac{\epsilon F}{m} \tag{78}$$

- The product ϵF is known as the *impulse* - describes the change in momentum.
- We see that a *unit impulse* ($\epsilon F = 1$) is equivalent to an initial velocity that is inversely proportional to the system mass, $\dot{x}_0 = 1/m$.

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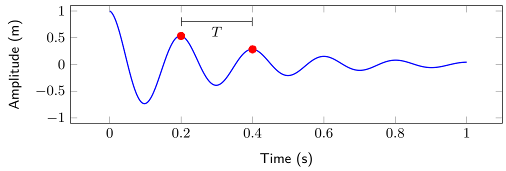
Estimating damping

Estimate damping - log-decrement method

- The amount of damping in a system is often unknown - it can be estimated from a response measurement $x(t)$.
- Assuming a viscous damping, the response decays exponentially,

$$x(t) = A\cos(\omega_d t - \phi)e^{-\zeta\omega_n t} \tag{79}$$

- Consider amplitude ratio after a period T.



Estimate damping - log-decrement method

- Consider amplitude ratio after a period T.

$$\frac{x_1}{x_2} = \frac{A\cos(\omega_d t_1 - \phi)e^{-\zeta\omega_n t_1}}{A\cos(\omega_d t_2 - \phi)e^{-\zeta\omega_n t_2}} \tag{80}$$

- Note that $t_2 = t_1 + T = t_1 + 2\pi/\omega_d$, and so,

$$\cos(\omega_d t_2 + \phi) = \cos(\omega_d(t_1 + 2\pi/\omega_d) + \phi) = \cos(\omega_d t_1 + 2\pi + \phi) = \cos(\omega_d t_1 + \phi) \tag{81}$$

- Note that the cos term does not change after T...

$$\frac{x_1}{x_2} = \frac{e^{-\zeta\omega_n t_1}}{e^{-\zeta\omega_n(t_1+T)}} = e^{\zeta\omega_n T} \tag{82}$$

- We define the so-called *log-decrement*, $\delta = \ln(x_1/x_2)$. This is something we can obtain from measurement.

Estimate damping - log-decrement method

- The log-decrement,

$$\delta = \ln(x_1/x_2) = \zeta\omega_n T = \zeta\omega_n\frac{2\pi}{\omega_d} = \frac{\zeta 2\pi}{\sqrt{1-\zeta^2}} \tag{83}$$

- Now we can solve for ζ to obtain,

$$\zeta = \frac{\delta}{\sqrt{(2\pi)^2 + \delta^2}} \tag{84}$$

- Note that for multiple periods,

$$\frac{x_1}{x_n} = \frac{x_1}{x_2}\frac{x_2}{x_3}\frac{x_3}{x_4}\cdots\frac{x_{n-1}}{x_n} = (e^{\zeta\omega_n T})^n = e^{n\zeta\omega_n T} \tag{85}$$

- The log-decrement (and so also ζ) can then be obtained using,

$$\delta = \frac{1}{n}\ln(x_1/x_2) \tag{86}$$

Summary

- Force relations for springs and dampers in series or parallel combinations.
- Derived the EoM for various mass-spring-damper type systems.
- Obtained the *homogenous* (unforced) solution for undamped and damped cases.
- Introduced the concept of *natural frequency of vibration*:
  - $\omega_n = \sqrt{k/m}$ is the undamped natural frequency.
  - $\omega_d = \sqrt{1-\zeta^2}\omega_n$ is the damped natural frequency, with ζ being the viscous damping factor.
- Exact solutions depends on IC, i.e. displacement and velocity at $t = 0$
- Viscous damping can be estimated using log-decrement method.

- Next time on POAV...
  - Forced response!